

Trend Following, Leveraged Re-Entry, and Volatility Decay

S&P 500 total return. One signal only: the 200-day moving average (the daily equivalent of Faber's 10-month rule). Tested as far back as the data allows.

Data. Total return throughout. Monthly S&P 500 total return back to 1901 (Shiller) for the Faber replication; a daily total return series from 1928 for everything else — real $\wedge SP500TR$ from 1988, and before that $\wedge GSFC$ price plus the Shiller dividend yield (this reconstruction tracks the real series with 0.50%/yr error and 0.9996 correlation over 1988–2026). Cash earns the 13-week T-bill ($\wedge IRX$; a 3.5% constant before 1960). The trend signal is lagged one day, so nothing uses information we could not have had.

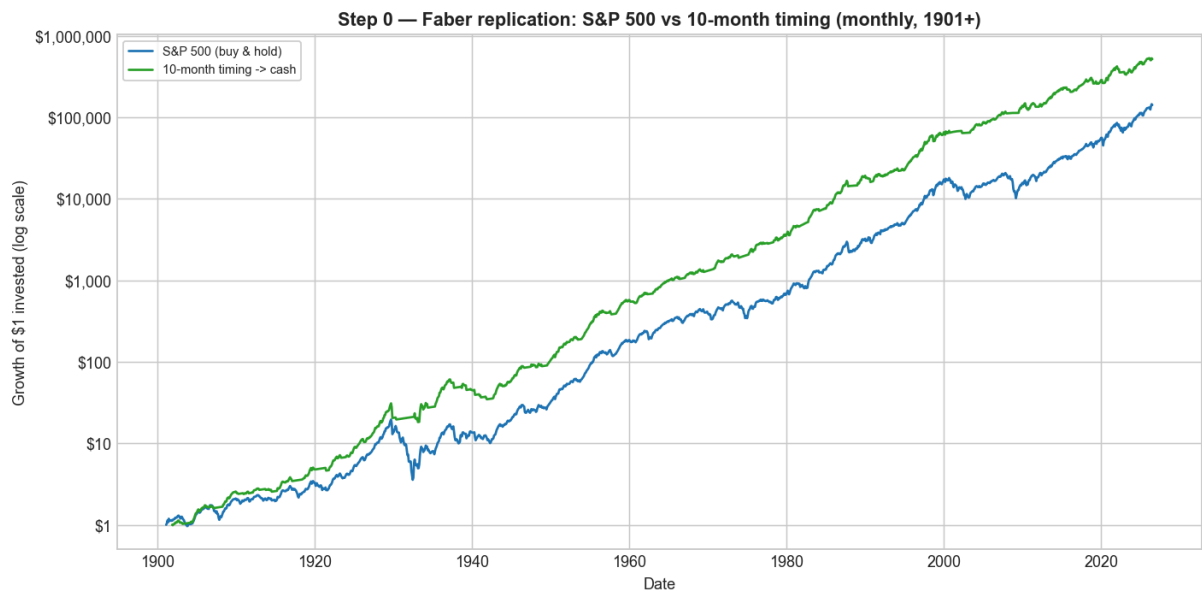
1. Buy & hold vs the Faber moving-average rule

The rule: hold the S&P 500 while it is **above** its moving average; move to **cash** while it is **below**. Faber uses the 10-month SMA on monthly data; we use that, then its daily twin (the 200-day SMA).

Monthly, 1901–2026 (Faber's exact setup):

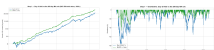
	S&P 500 buy & hold	10-month timing → cash
CAGR	9.95%	11.24%
Volatility	15.4%	10.8%
Sharpe	0.44	0.69
Max drawdown	−81.8%	−43.0%

(Faber's published drawdowns are −83.66% → −42.24%; we reproduce them to within a point.)



Daily, 200-day SMA, 1928–2026:

	Buy & Hold 1×	MA200 → Cash
CAGR	10.14%	11.29%
Volatility	18.9%	12.6%
Sharpe	0.40	0.60
Sortino	0.56	0.84
Max drawdown	−83.9%	−46.2%
Calmar	0.12	0.24



The moving-average rule keeps essentially all of the return while cutting volatility by a third and halving the worst drawdown, so its Sharpe, Sortino and Calmar are all much higher. The 200-day MA adds clear risk-adjusted value.

2. Leverage returns

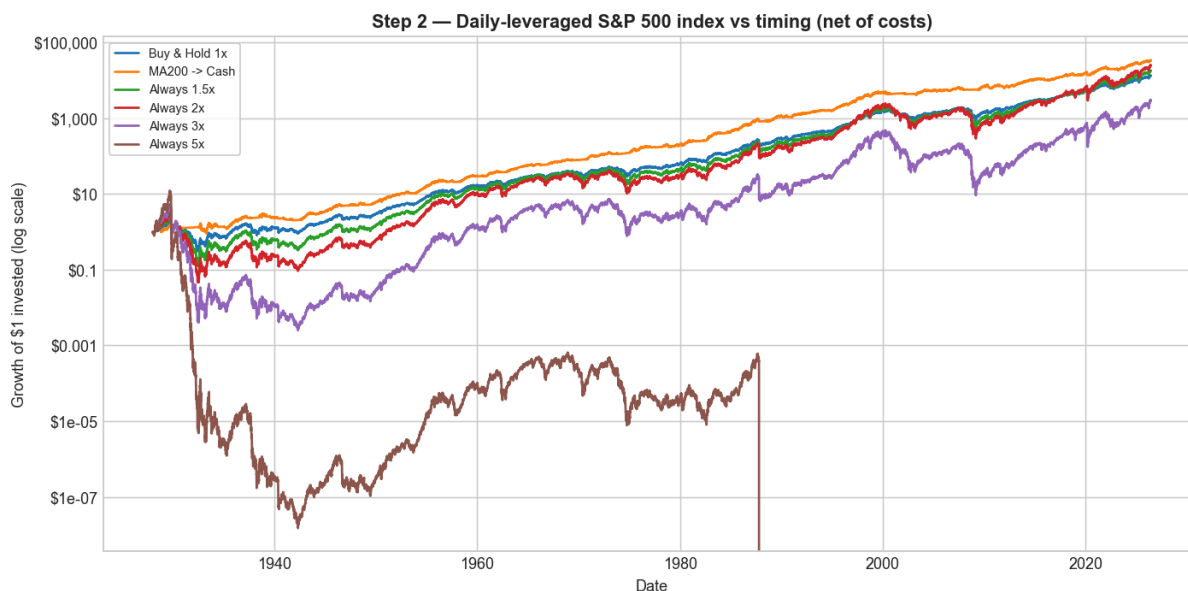
A daily **Lx leveraged** return is simply that day's S&P 500 total return multiplied by L, and then compounded day by day:

$$\text{leveraged_return}[t] = L \times \text{sp500_return}[t] \quad (\text{before fees / financing})$$

For example, the 2x series is just the daily S&P total return $\times 2$, compounded. Borrowed money costs the financing rate, and leveraged funds charge a fee ($\sim 0.9\%$ /yr); both are included below.

Holding **constant** daily leverage on the index, net of costs:

	CAGR	Grew \$1 to
1x (buy & hold)	10.14%	\$13,021
Always 1.5x	10.46%	\$17,378
Always 2.5x	10.15%	\$13,110
Always 3x	8.43%	\$2,803



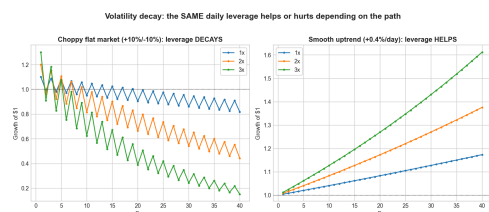
Constant leverage barely helps at 1.5x and *loses ground* by 3x — across a full century (including 1929 and 2008), constant 3x ends below plain buy & hold and far below the moving-average rule.

3. Volatility decay, and buying leverage at the lows

Volatility decay. A +10% day followed by a -10% day:

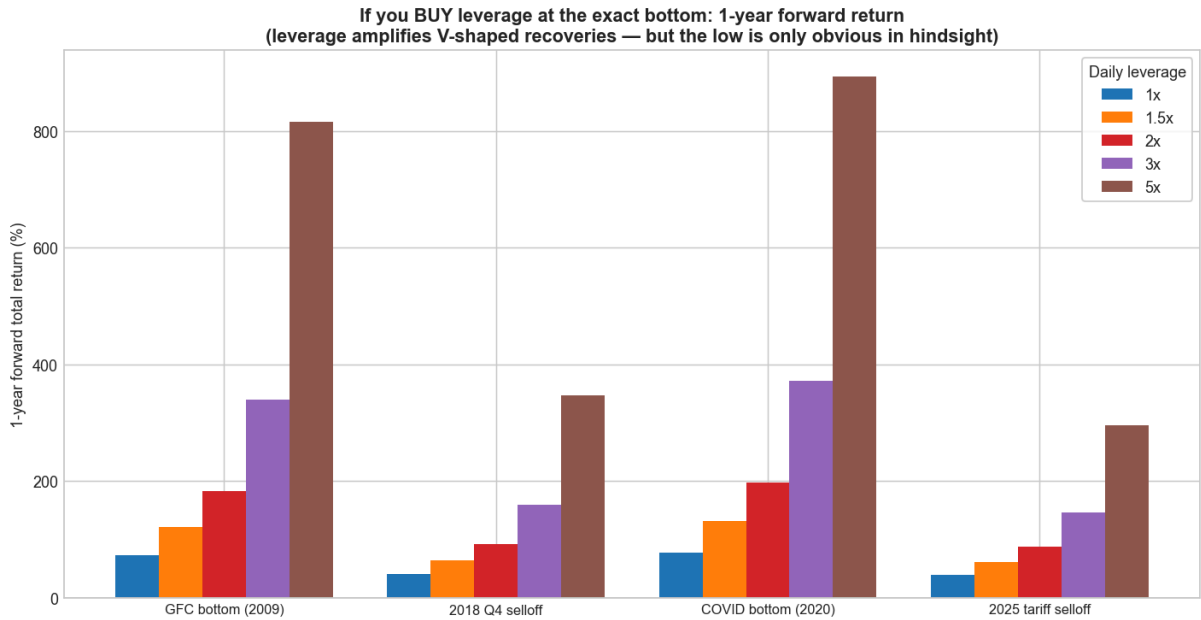
	1x	2x	3x
Two-day return	-1.0%	-4.0%	-9.0%

The market round-trips to roughly flat, but leverage loses — and the loss grows with the *square* of leverage. The annual penalty is $\approx \frac{1}{2} \cdot L^2 \cdot \sigma^2$: at 20% volatility, $\sim 2\%$ /yr for 1x, **8%/yr for 2x**, **18%/yr for 3x**.



But decay only bites in choppy/falling markets. In a one-directional rally — like the rebound off a market bottom — leverage amplifies the gain. Forward **1-year** total return if you had bought at the exact low:

Bottom	1×	1.5×	2×	3×
GFC (2009-03-09)	+72%	+122%	+182%	+339%
2018 Q4 (2018-12-24)	+40%	+64%	+92%	+159%
COVID (2020-03-23)	+78%	+132%	+198%	+372%
2025 tariff selloff (2025-04-08)	+39%	+61%	+87%	+146%



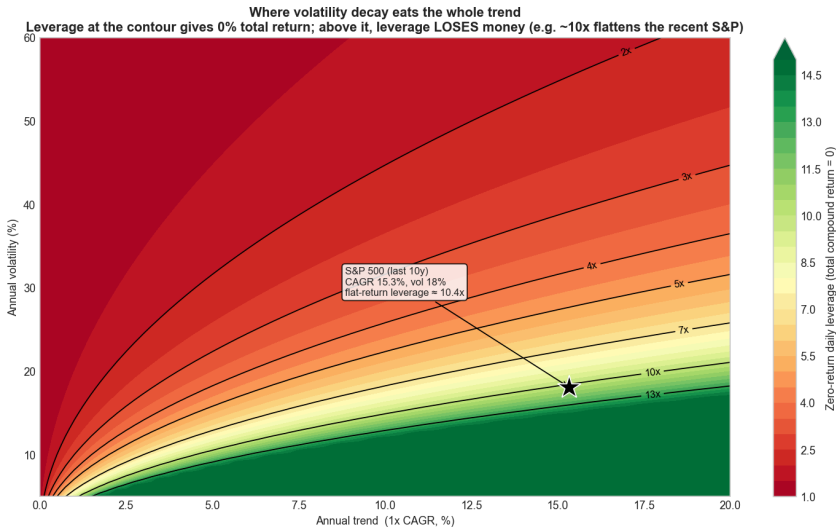
Leverage can work strongly in your favour **if you time it** — buying into the recovery off a low. (The low, of course, is only obvious in hindsight.)

4. The leverage at which the total return goes flat

For a given trend (the 1× CAGR g) and volatility σ , the compound return of $L\times$ leverage is $L \cdot \mu - \frac{1}{2} \cdot L^2 \cdot \sigma^2$. Setting it to zero gives the leverage at which volatility decay exactly cancels the trend, so the **total return is flat (0%)**:

$$L_{\text{zero}} = 2 \cdot g / \sigma^2 + 1$$

Below it, leverage still grows; above it, leverage **loses money**. The map below plots L_{zero} for every combination of trend and volatility. For the S&P over the **last 10 years** (CAGR 15.3%, volatility 18.1%) the flat point is **≈ 10×** — so a hypothetical "10× S&P" would have gone essentially nowhere despite a strong decade, while higher still would have bled toward zero.

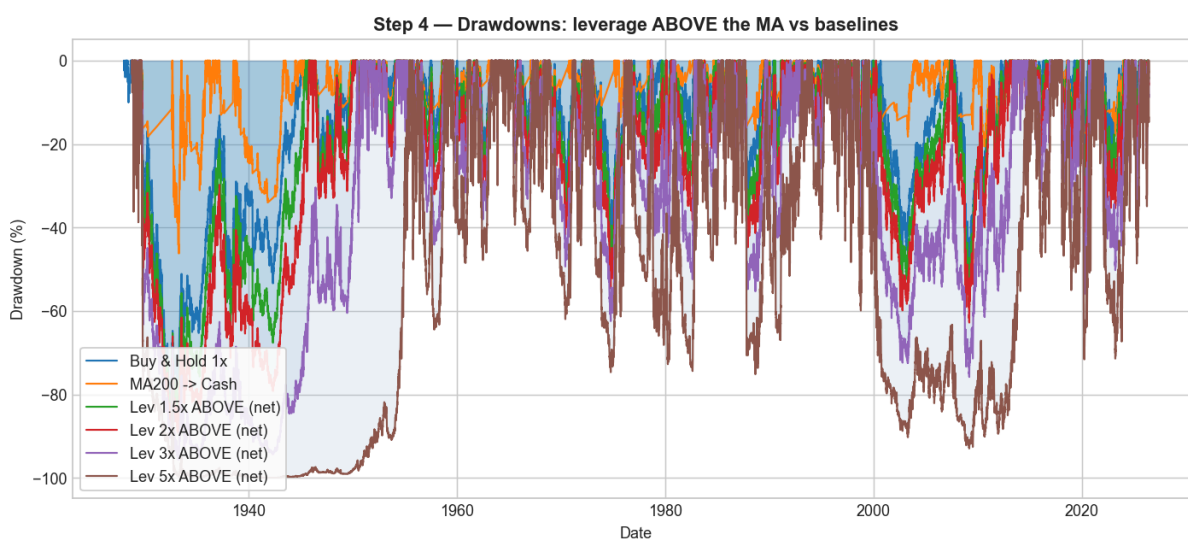
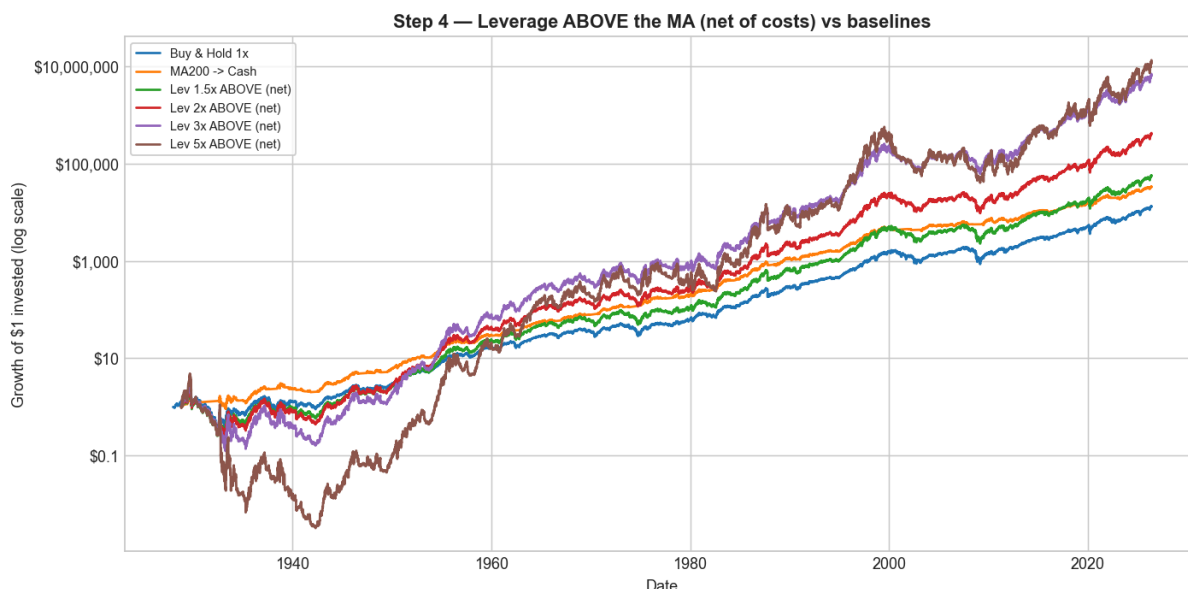


(For reference, the leverage that merely *ties* 1× buy & hold is lower — $\approx 3.1\times$ for the S&P — and the growth-maximising level is $\approx 2\times$; see [charts/F3_breakeven_leverage_map.png](#) and [charts/F3_optimal_leverage_curve.png](#).)

5. The switching strategy: leverage the uptrend

Use the 200-day MA to switch between leveraged and ordinary exposure: **hold $L\times$ leverage while the market is ABOVE the MA** — the calm, rising regime where the kind of one-directional gains seen in §3 actually happen — and drop back to plain 1× while it is below. Daily total return, 1928–2026, net of costs:

Strategy	Grew \$1 to	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar
Buy & Hold 1×	\$13,021	10.1%	18.9%	0.40	0.56	−83.9%	0.12
MA200 → Cash	\$33,090	11.3%	12.6%	0.60	0.84	−46.2%	0.24
Lev 1.5× above MA	\$55,471	11.9%	23.6%	0.43	0.60	−85.7%	0.14
Lev 2× above MA	\$402,863	14.2%	28.9%	0.47	0.66	−89.2%	0.16
Lev 3× above MA	\$6,429,403	17.5%	40.4%	0.50	0.71	−95.7%	0.18
Lev 5× above MA	\$11,648,321	18.2%	64.7%	0.53	0.74	−99.9%	0.18

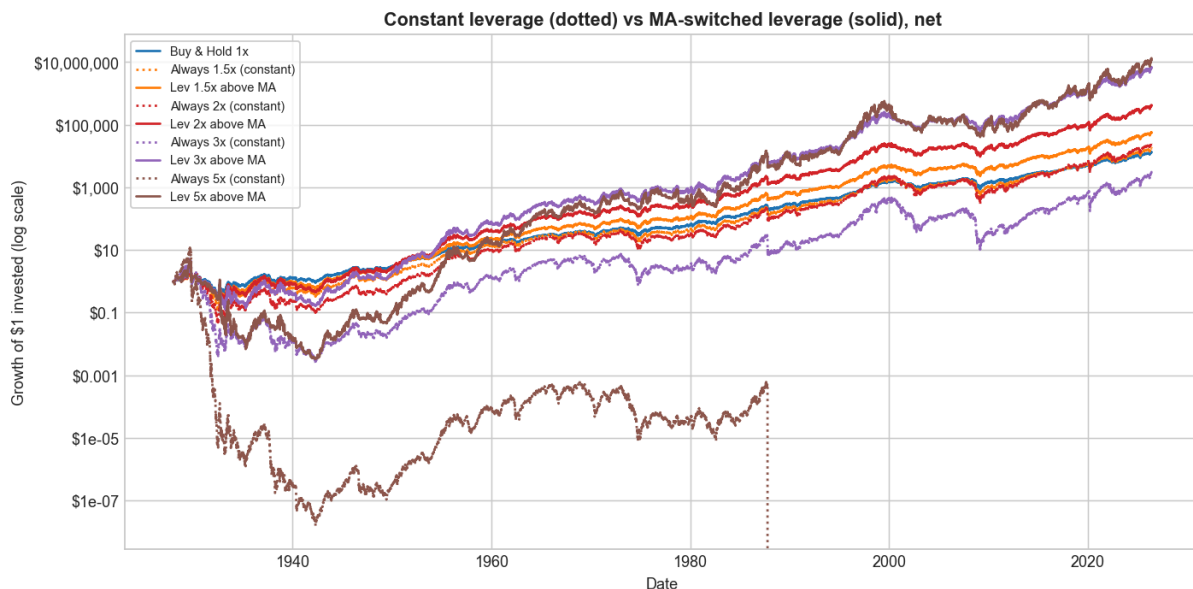


Leveraging the uptrend gets **better** as leverage rises — \$1 grows to \$402,863 at 2× and \$11.6M at 5× (versus \$13,021 for buy & hold) — and it beats buy & hold on CAGR, Sharpe, Sortino and Calmar at every level. The trade-offs: maximum drawdown deepens with leverage (−86% at 1.5× to −99.9% at 5×, because you are leveraged going *into* fast crashes), and it still does not beat the plain move-to-cash rule on a risk-adjusted basis.

6. Does the MA switch add value over just holding leverage?

Is the switching doing the work, or could you just hold constant leverage? Holding each leverage level *constantly* (dotted) versus only **above the MA** (solid), 1928–2026, net:

Strategy	Grew \$1 to	CAGR	Vol	Sharpe	Max DD
Buy & Hold 1×	\$13,021	10.1%	18.9%	0.40	−84%
Always 1.5×	\$17,378	10.5%	28.4%	0.35	−95%
Lev 1.5× above MA	\$55,471	11.9%	23.6%	0.43	−86%
Always 2×	\$23,866	10.8%	37.8%	0.36	−99%
Lev 2× above MA	\$402,863	14.2%	28.9%	0.47	−89%
Always 3×	\$2,803	8.4%	56.8%	0.36	−100%
Lev 3× above MA	\$6,429,403	17.5%	40.4%	0.50	−96%
Always 5×	\$0 (wiped out)	—	94.6%	0.36	−100%
Lev 5× above MA	\$11,648,321	18.2%	64.7%	0.53	−99.9%

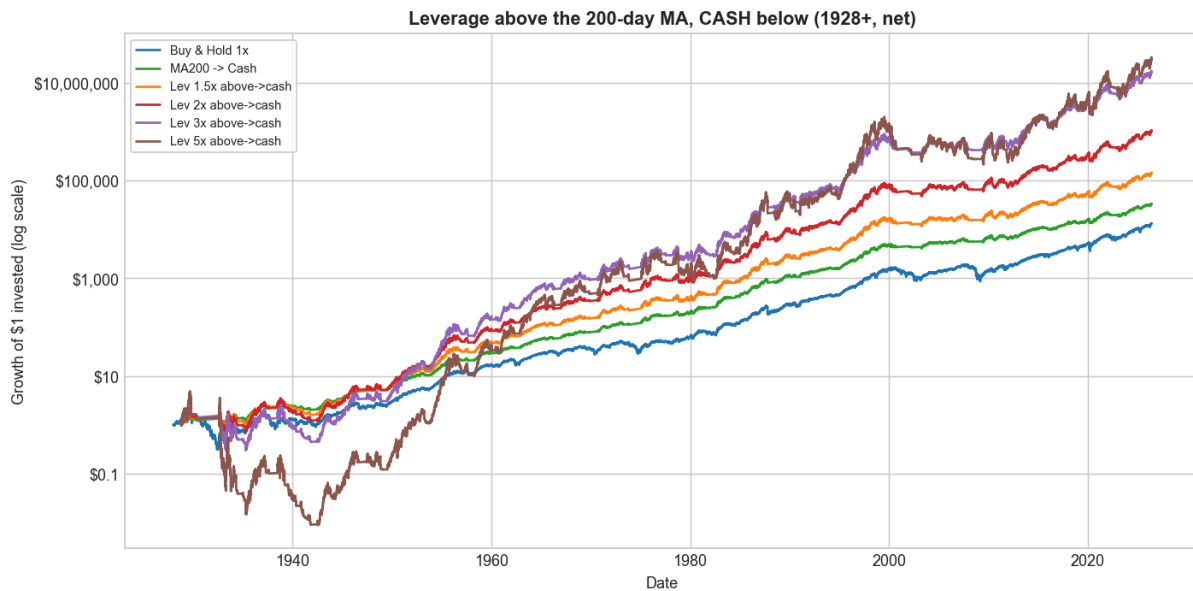


The switch adds enormous value: at every level, the MA-switched version has higher CAGR, higher Sharpe **and** a shallower drawdown than holding the same leverage all the time. Constant 5× was **wiped out** (a single −22% day in 1987 at 5× is a −110% move); the switched 5× grew \$1 to \$11.6 million. Constant leverage just piles risk on with little extra Sharpe; the trend filter is what makes leverage pay.

7. Leverage → cash: sidestep the downturns

Instead of dropping to plain 1× below the MA, go all the way to **cash** — so the strategy is leveraged in uptrends and completely out during below-trend slumps. 1928–2026, net:

Strategy	Grew \$1 to	CAGR	Vol	Sharpe	Max DD	Calmar
Buy & Hold 1×	\$13,021	10.1%	18.9%	0.40	−84%	0.12
MA200 → Cash	\$33,090	11.3%	12.6%	0.60	−46%	0.24
Lev 1.5×	\$140,934	13.0%	18.9%	0.53	−63%	0.21
Lev 2×	\$1,023,603	15.3%	25.3%	0.53	−75%	0.20
Lev 3×	\$16,337,433	18.6%	37.9%	0.54	−90%	0.21
Lev 5×	\$29,600,903	19.3%	63.1%	0.54	−99.8%	0.19

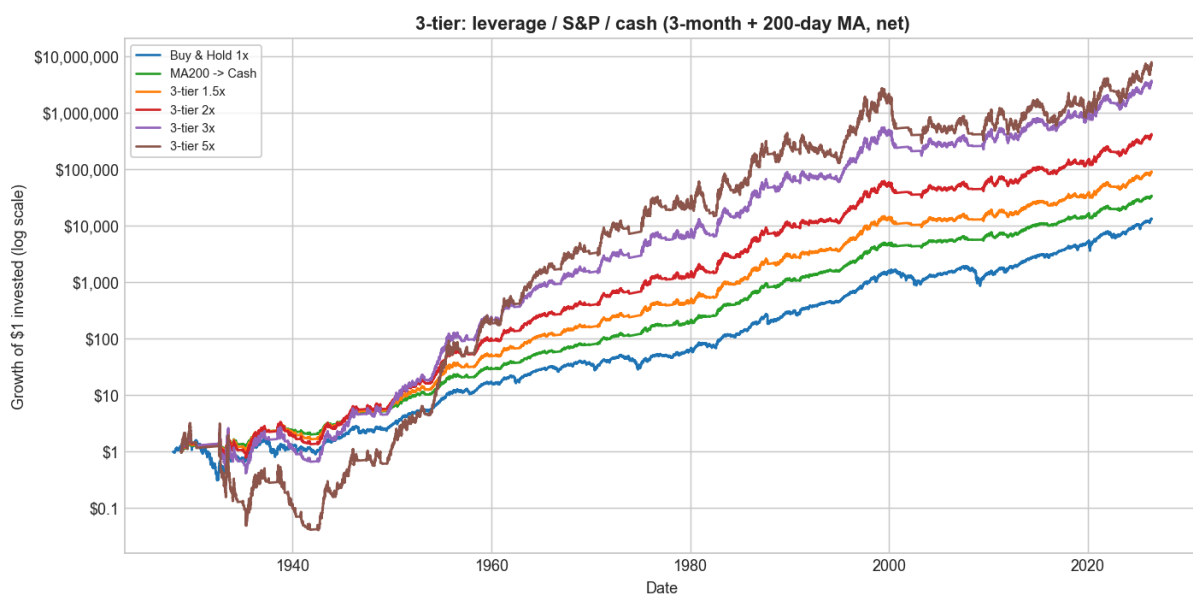


Going to cash (not 1×) below the trend both **raises returns** and **shallows the drawdown** versus leverage-to-1×: e.g. 2× above→cash has a −75% drawdown at Sharpe 0.53, against the leverage-to-1× version's −89% at 0.47. Sharpe sits near 0.53–0.54 at every leverage (leverage scales the in-market return and risk together), so you simply dial CAGR up or down with leverage. It still trails plain MA→cash on Sharpe (0.60) and drawdown — the price of the much higher compound return.

8. A three-tier rule: leverage → S&P → cash

A finer version uses a fast (3-month) MA *and* the slow (200-day) MA: **leverage** above both, **plain 1× S&P** on a mild pullback (below the 3-month but above the 200-day), **cash** below the 200-day. 1928–2026, net:

Strategy	Grew \$1 to	CAGR	Vol	Sharpe	Max DD	Calmar
Buy & Hold 1×	\$13,021	10.1%	18.9%	0.40	−84%	0.12
MA200 → Cash	\$33,090	11.3%	12.6%	0.60	−46%	0.24
3-tier 1.5×	\$87,949	12.4%	17.4%	0.53	−58%	0.21
3-tier 2×	\$398,471	14.2%	22.5%	0.53	−68%	0.21
3-tier 3×	\$3,426,219	16.7%	33.0%	0.52	−84%	0.20
3-tier 5×	\$6,868,411	17.6%	54.2%	0.50	−99%	0.18



The middle "plain S&P" tier trims volatility and drawdown versus the straight leverage→cash rule at low leverage (3-tier 1.5×: vol 17.4%, drawdown −58%, Calmar 0.21). Of all the leveraged variants, the 3-tier rule's risk-adjusted numbers

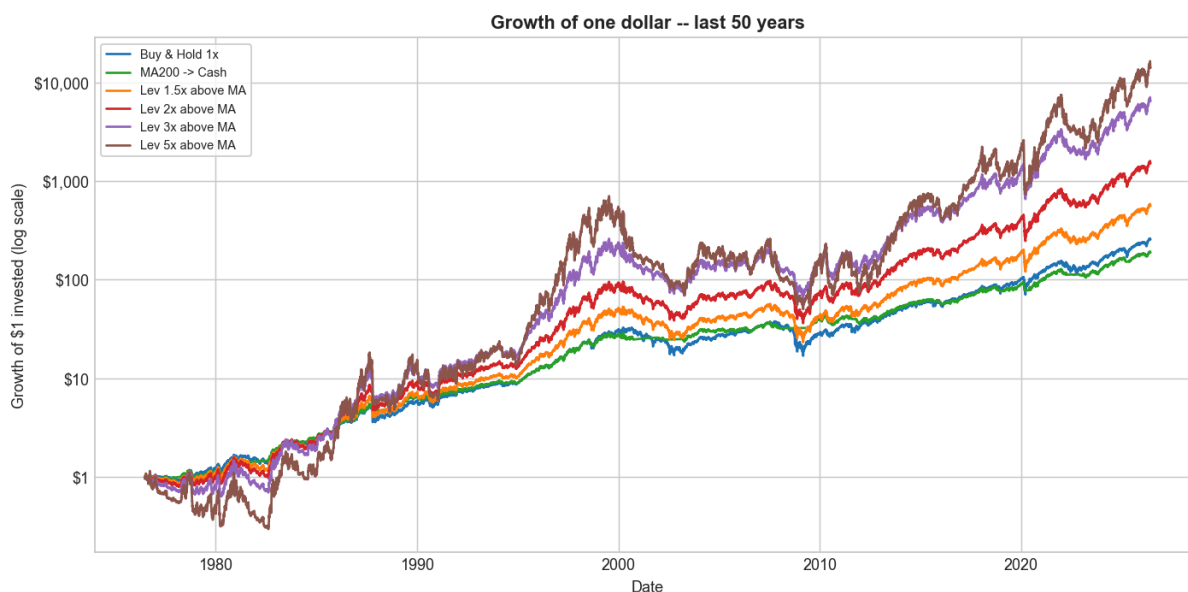
come closest to the plain MA→cash rule while keeping a much higher compound return.

9. Performance over recent horizons (last 50 / 30 / 15 years)

The 1928-start drawdowns are dominated by 1929/1987. Here is the leverage-the- uptrend strategy over more recent windows (net of costs); volatility is shown so the Sharpe can be checked directly.

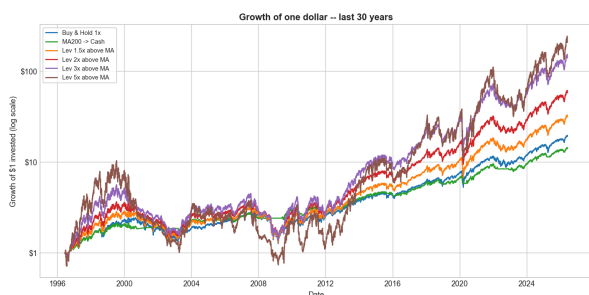
Last 50 years:

Strategy	Grew \$1 to	CAGR	Vol	Sharpe	Max DD	Calmar
Buy & Hold 1×	\$257	11.7%	17.4%	0.49	−55%	0.21
MA200 → Cash	\$189	11.1%	11.7%	0.60	−21%	0.54
Lev 1.5× above MA	\$567	13.5%	21.8%	0.50	−59%	0.23
Lev 2× above MA	\$1,533	15.8%	26.7%	0.53	−63%	0.25
Lev 3× above MA	\$6,623	19.2%	37.4%	0.55	−76%	0.25
Lev 5× above MA	\$14,644	21.2%	60.0%	0.56	−93%	0.23



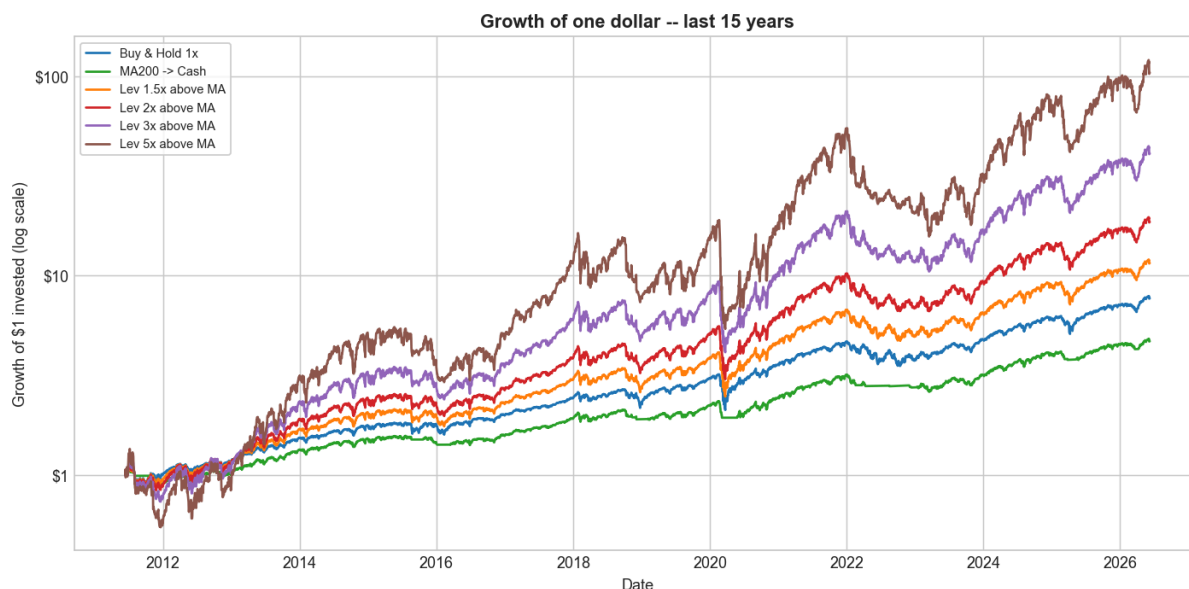
Last 30 years:

Strategy	Grew \$1 to	CAGR	Vol	Sharpe	Max DD	Calmar
Buy & Hold 1×	\$19	10.4%	19.1%	0.50	−55%	0.19
MA200 → Cash	\$14	9.2%	12.1%	0.61	−21%	0.45
Lev 1.5× above MA	\$32	12.2%	23.4%	0.52	−59%	0.21
Lev 2× above MA	\$58	14.5%	28.4%	0.54	−63%	0.23
Lev 3× above MA	\$141	18.0%	39.2%	0.56	−76%	0.24
Lev 5× above MA	\$209	19.5%	62.3%	0.57	−93%	0.21



Last 15 years:

Strategy	Grew \$1 to	CAGR	Vol	Sharpe	Max DD	Calmar
Buy & Hold 1×	\$7.7	14.6%	17.3%	0.79	−34%	0.43
MA200 → Cash	\$4.7	10.9%	11.8%	0.81	−18%	0.61
Lev 1.5× above MA	\$11.5	17.8%	21.7%	0.79	−40%	0.44
Lev 2× above MA	\$18.6	21.6%	26.7%	0.81	−46%	0.47
Lev 3× above MA	\$41.1	28.2%	37.5%	0.81	−56%	0.50
Lev 5× above MA	\$104.7	36.5%	60.1%	0.80	−72%	0.51



Across all three windows the leverage-the-uptrend strategy earns far higher compound returns at **roughly the same Sharpe** as buy & hold (over the last 15 years, 5× grew \$1 to ~\$105 at Sharpe 0.80, versus buy & hold's \$7.7 at 0.79). The plain MA→cash rule keeps the best Sharpe, Calmar and drawdown throughout, but the lowest dollar growth.

Educational research only — not investment advice. Reproduce with `python run_faber_leverage.py`; figures are in `charts/`, numbers in `results/`.